YANSHU ZHANG

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RESEARCH INTERESTS

I build interactive virtual worlds from images and videos, spanning point-based neural rendering, feed-forward 3D reconstruction and novel view synthesis, video-to-4D, and 3D editing and animation.

EDUCATION

Simon Fraser University *09/2021 - Present*
Ph.D. in Computer Science, supervised by Prof. Ke Li

University of Science and Technology of China *09/2017 - 07/2021*
B.S. in Computer Science

PUBLICATIONS

** denotes equal contribution*

PointWeave: Feed-Forward View Synthesis with Explicit Point Interface
Yanshu Zhang, Shichong Peng, Chirag Vashist, Ke Li
Under Submission · [Project] [Paper]

RigPAPR: Rig-Based Animation of Static Neural Point Clouds from a Single-View Video
Shichong Peng, **Yanshu Zhang**, Ke Li
NeurIPS 2026 · [arXiv]

The Fault in Our Metrics: Revisiting Generative Model Evaluation with Chamfer Distance
Chirag Vashist, Vaibhav Santhosh, Tristan Engst, **Yanshu Zhang**, Ke Li
NeurIPS 2026 (Evaluations & Datasets Track)

PointGT: Simultaneous Geometry and Texture Editing for Point-Based Representations
Yanshu Zhang, George Shramko, Pratul P. Srinivasan, Ke Li
ECCV 2026 · [Project] [arXiv] [Code]

P-CORE: Self-Supervised Surface Consistency for Point-Based Neural Editing
Yanshu Zhang, Shichong Peng, Mehran Aghabozorgi, Alireza Moazeni, Ke Li
ECCV 2026 · [Project] [arXiv] [Code]

Tackling Misattribution in 3D Intrinsic Decomposition via Proximity Attention Point Rendering
Alireza Moazeni, Shichong Peng, **Yanshu Zhang**, Chirag Vashist, Ke Li
ECCV 2026 · [arXiv]

PAPR Up-close: Close-up Neural Point Rendering without Holes
Yanshu Zhang, Chirag Vashist, Shichong Peng, Ke Li
3DV 2026 · [Project] [Paper]

WIMLE: Uncertainty-Aware World Models with IMLE
Mehran Aghabozorgi, Alireza Moazeni, **Yanshu Zhang**, Ke Li
ICLR 2026 · [Project] [arXiv] [Code]

PROSE: Point Rendering of Sparse-Controlled Edits to Static Scenes
Yanshu Zhang, Shichong Peng, Mehran Aghabozorgi, Alireza Moazeni, Ke Li
ECCV 2024 Workshop

PAPR in Motion: Seamless Point-level 3D Scene Interpolation
Shichong Peng, **Yanshu Zhang**, Ke Li
*CVPR 2024 (**Highlight**)* · [Project] [arXiv] [Code]

PAPR: Proximity Attention Point Rendering
Yanshu Zhang*, Shichong Peng*, Alireza Moazeni, Ke Li
*NeurIPS 2023 (**Spotlight**)* · [Project] [arXiv] [Code]

ACADEMIC SERVICE

Reviewer for CVPR, ECCV, ICCV, 3DV, SIGGRAPH, SIGGRAPH Asia, ICML, ICLR, NeurIPS, TPAMI

PATENTS

Method and System for Accelerated Operation of Layers Used in a Machine Learning Model and Differentiable Point Rendering Using Proximity Attention
Yanshu Zhang, Yuzhen Mao, Shichong Peng, Alireza Moazeni, Ke Li
CA Patent Application: 3292617 (2024)

EXPERIENCE

National University of Singapore *01/2021 - 05/2021*
Research Exchange Programme

TECHNICAL SKILLS

Programming: Python, C++, CUDA (custom kernels)
Frameworks & Tools: PyTorch, multi-GPU distributed training (DDP), SLURM clusters
Research Areas: Neural rendering, NeRF / 3D Gaussian Splatting, point-based rendering, feed-forward 3D reconstruction, novel view synthesis, video-to-4D, 3D editing and animation, generative models

AWARDS

SFU Ph.D. Research Scholarship *(2022, 2024)*
SFU Graduate Fellowship *(2023, 2025)*
Silver Award for Outstanding Students of USTC *(2019)*
Bronze Award for Outstanding Students of USTC *(2018)*

TEACHING EXPERIENCE

Generative Models, Teaching Assistant, SFU *(2025)*
Machine Learning, Teaching Assistant, SFU *(2022)*